

FRE-GT-9743 Assignment 1

Bond Forward Business

Q1. Client motivation, key terms, and settlement

A client may enter a long bond forward to lock in today the price of a Treasury bond that it plans to buy at the settlement date T_s . This hedges the risk that the bond price rises before T_s , for example because yields fall.

Important contract terms include the notional amount, the specific Treasury bond, settlement date T_s , maturity T_m , forward strike K , coupon schedule, settlement convention, and CSA/collateral terms.

At T_s , Bank A delivers the Treasury bond and the client pays the agreed delivery price K . The value to the long side is

$$V(T_s) = B(T_s; T_s, T_m) - K.$$

Q2. Why buy the spot bond at inception, and how is it funded?

Bank A is short the bond forward and must deliver the Treasury at T_s . If it waits until T_s to buy the bond, it is exposed to changes in the bond price. Buying the spot bond at $t = 0$ locks in the asset needed for future delivery and hedges the main bond-price risk.

The spot bond purchase requires cash. The bank's treasury/funding function provides or coordinates the initial funding, while the repo desk finances the bond on a secured basis. The Treasury is pledged as collateral and the repo desk provides cash against it. Since no haircut is specified, the bond position is treated as fully repo-funded.

Q3. Forward bond price from spot price, repo rate, and coupons

Yes. Let

$$B_0 = B(0; 0, T_m).$$

If the bond is financed through repo at a constant simple annual rate R , the initial purchase cost grows to

$$B_0(1 + RT_s).$$

at the settlement date T_s .

If a coupon C_i is paid at time $t_i < T_s$, it is used to reduce the outstanding repo balance. Therefore, the coupon reduces the amount owed at T_s by

$$C_i[1 + R(T_s - t_i)].$$

Hence, using the spot bond price, repo rate, and coupon schedule,

$$B(0; T_s, T_m) = B_0(1 + RT_s) - \sum_{t_i < T_s} C_i[1 + R(T_s - t_i)].$$

Thus, $B(0; T_s, T_m)$ can be computed directly from $B(0; 0, T_m)$, the repo rate R , and the coupons paid before T_s .

Q4. MTM at an intermediate time t

For any time after inception and up to the settlement date, the current MTM is

$$V(t) = df^{CSA}(t, T_s) [B(t; T_s, T_m) - K].$$

Here, $B(t; T_s, T_m)$ is the current forward bond price for settlement at T_s , while K is the strike fixed at inception. The difference is discounted from T_s back to t using the CSA discount factor.

Q5. Market risk and desk revenue

Following Q2-Q3, the desk largely hedges the main directional bond-price risk because it already owns the bond required for delivery and has arranged its financing.

This does not imply zero revenue. The desk can earn a bid-ask or funding spread by quoting the client a forward price that embeds a repo/funding rate above its actual financing cost. Revenue therefore comes from spread and pricing, rather than from leaving the bond-price exposure unhedged.

***Q5. Relation to risk-neutral pricing**

Q2-Q3 form a cash-and-carry replication. The desk buys the spot Treasury, finances it through repo, and uses intermediate coupons to reduce the financing balance. By no-arbitrage, the forward price must equal the cost of this replicating strategy.

Under consistent assumptions, martingale/risk-neutral pricing gives the same result. In the collateralised framework, the underlying is carried at the secured funding rate; here the repo rate R is the relevant secured carry rate. Hence,

$$\text{replication price} = \text{risk-neutral price}.$$

Equivalently, under the appropriate T_s -forward measure, the fair forward price can be written as the conditional expectation of the bond price at T_s .